

KIMBERLEY, South AfricaWMO: **684380**Lat: **28.8061S**Lon: **24.7697E**Elev: **1196**StdP: **87.76**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.1	-0.6	-15.2	1.2	14.6	-12.9	1.4	14.0	11.1	15.5	10.1	16.8	1.2	10	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	36.2	16.5	35.0	16.4	33.9	16.4	21.1	28.2	20.4	27.4	19.7	26.7	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	16.3	23.0	18.6	15.5	22.5	17.8	14.8	22.2	67.5	27.7	64.6	27.4	62.2	26.7	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.3	9.1	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-5.5	38.0	1.4	1.1	-6.5	38.8	-7.3	39.4	-8.1	40.0	-9.1	40.8
			-6.7	22.9	1.2	2.8	-7.6	24.9	-8.4	26.5	-9.1	28.1	-10.0	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	24.8	24.4	22.3	18.0	14.2	10.8	10.5	13.3	17.1	20.4	22.3	24.0
	DBStd	6.12	2.68	2.69	3.06	3.46	3.45	3.22	3.57	4.09	4.26	3.79	3.70	2.79
	HDD10.0	94	0	0	0	1	8	29	38	16	2	0	0	0
	HDD18.3	925	1	1	6	47	132	227	243	162	72	24	11	1
	CDD10.0	3182	459	402	381	240	138	52	54	118	215	321	369	433
	CDD18.3	970	201	170	128	36	3	0	0	6	35	87	129	176
	CDH23.3	11694	2261	1782	1325	423	102	7	18	148	616	1211	1691	2111
(13)	CDH26.7	5348	1147	857	566	104	7	0	1	20	223	528	824	1071
(14) Wind	WSAvg	3.6	4.1	3.7	3.4	2.9	2.8	2.9	3.1	3.6	3.8	4.1	4.3	4.3
(15) Precipitation	PrecAvg	406	65	70	63	42	16	8	5	7	11	32	42	48
	PrecMax	763	258	332	154	159	81	69	40	69	70	161	143	161
	PrecMin	186	1	5	3	0	0	0	0	0	0	0	5	1
	PrecStd	142	51	54	37	36	20	12	10	13	17	32	34	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	36.5	35.4	32.1	27.8	24.2	25.4	28.8	33.4	36.0	36.7	38.1
		MCWB	16.9	17.9	17.0	16.2	13.3	11.9	11.1	12.4	14.6	15.8	15.7	16.5
	2%	DB	35.9	35.0	33.4	29.1	25.9	22.2	22.9	26.8	31.3	33.5	34.9	35.9
		MCWB	16.6	17.2	17.1	15.3	12.5	10.9	10.1	11.6	13.8	14.8	15.8	16.1
	5%	DB	34.2	33.6	31.8	27.2	24.4	21.0	21.2	24.9	29.2	31.9	33.5	34.1
		MCWB	16.8	17.6	16.9	15.1	12.3	10.2	9.6	11.1	12.7	14.6	15.5	16.1
	10%	DB	32.8	31.9	30.0	25.8	22.7	19.3	19.5	22.9	27.1	29.9	31.6	32.3
		MCWB	17.2	17.8	16.8	14.7	12.2	9.5	9.1	10.6	12.1	14.1	15.1	16.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	23.7	21.2	19.0	16.1	13.2	12.6	14.7	16.7	19.6	20.8	20.8
		MCDB	29.2	28.1	27.3	25.4	22.5	21.0	22.2	24.1	28.0	25.7	26.7	28.7
	2%	WB	20.8	21.5	20.2	17.8	14.5	12.1	11.3	13.2	15.3	17.8	19.1	19.8
		MCDB	27.9	28.3	26.4	24.7	21.6	19.8	20.6	22.9	26.1	27.1	26.7	27.1
	5%	WB	20.2	20.5	19.3	17.0	13.6	11.1	10.3	12.2	14.3	16.7	18.1	19.1
		MCDB	27.1	27.8	26.2	23.5	21.2	18.9	19.5	21.9	25.4	26.6	26.3	26.7
	10%	WB	19.4	19.7	18.5	16.2	12.8	10.2	9.4	11.0	13.3	15.7	17.2	18.3
		MCDB	26.5	26.8	25.6	22.3	20.4	17.8	18.4	21.1	24.3	25.5	26.0	26.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	13.8	13.9	14.3	15.6	16.2	16.7	17.1	17.4	16.6	16.1	15.4
		MCDBR	16.9	16.2	16.4	15.6	17.5	17.7	18.0	18.7	19.4	18.7	18.2	18.0
		MCWBR	5.8	5.7	5.8	6.1	7.9	9.1	9.1	9.2	8.9	7.7	7.2	7.0
	5% WB	MCDBR	12.8	12.8	12.8	12.7	14.6	15.5	16.4	16.4	16.7	15.4	14.3	14.2
		MCWBR	4.9	5.4	5.0	5.8	7.3	8.4	8.9	8.6	8.2	6.8	6.3	5.8
(40) Clear-Sky Solar Irradiance	taub		0.340	0.349	0.340	0.307	0.284	0.267	0.265	0.310	0.352	0.363	0.322	0.326
	taud		2.475	2.450	2.452	2.520	2.562	2.579	2.566	2.418	2.291	2.294	2.469	2.497
	Ebn at Noon		1002	974	951	937	916	914	930	918	923	949	1012	1017
	Edh at Noon		118	117	111	93	80	74	78	101	128	136	118	116
(44) All-Sky Solar Radiation	RadAvg		7.51	6.78	5.99	4.87	4.08	3.65	3.98	4.82	5.97	6.96	7.79	7.99
	RadStd		0.51	0.57	0.32	0.25	0.18	0.16	0.12	0.23	0.26	0.32	0.39	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.68	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page